

Vol 11 Architectural Scaffold v0.1 — Generative Geometry & Topology (BST Mathematical Foundations)

Filed: 2026-05-23 Saturday (Lyra, Wave 2 scaffold; deep-NT thread) Companion to: INDEX.md

Anchoring BST results (existing coverage by chapter)

Ch 1 Bounded Hermitian Symmetric Domains (~70%)

Anchors: Cartan 1894 + Helgason 1978 X.6.1 + Vol 0 Ch 1 + Vol 0 Ch 9 - Cartan classification: $4 + 2 = 6$ types of irreducible bounded HSDs - Helgason 1978 Theorem X.6.1 (type IV unique infinite family with rank=2 at all $\dim \geq 2$) - Strong-Uniqueness Theorem (Paper #125 v0.10.5 FORMAL) selects D_{IV}^5 uniquely

Ch 2 Bergman Reproducing Kernels (~75%)

Anchors: Bergman 1922 + Faraut-Koranyi 1994 + T2442 - Bergman 1922 reproducing-kernel construction - Faraut-Koranyi 1994 explicit kernel formula on type IV domains - T2442 $c_{FK} \cdot \pi^{(9/2)} = 225$ EXACT (Strong-Uniqueness C13)

Ch 3 Wallach K-Type Representation Theory (~70%)

Anchors: Wallach 1976 + Vol 1 Ch 2 + Ch 5 - Wallach 1976 classification of holomorphic discrete series on bounded HSDs - K-type structure of Bergman $H^2(D_{IV}^5)$ - Wallach's contribution to BST Strong-Uniqueness C8 (lowest K-type Casimir = $C_2 = 6$)

Ch 4 Holomorphic Discrete Series + Highest Weights (~60%)

Anchors: Standard Lie-theory machinery + Wallach 1976 + Cartan classification - Standard Lie-theory representation theory of semisimple groups - BST cross-link: Casimir algebra Vol 1 Ch 5

Ch 5 D_{IV}^5 Geometry: Coset + Boundary + Coordinates (~75%)

Anchors: Cartan 1894 + Hua 1958 + Vol 0 Ch 1 + Ch 4 + Vol 1 Ch 2 - Type IV bounded domain realization (Cartan 1894 + Hua 1958) - Shilov boundary structure (conformal boundary of D_{IV}^5) - Symmetric-pair decomposition $\mathfrak{so}(5,2) = (\mathfrak{so}(5) \oplus \mathfrak{so}(2)) \oplus \mathfrak{m}$

Ch 6 K3 Surfaces + Niemeier Lattices + Leech (~60%)

Anchors: Hodge K3 1962/64 + Mathieu 1861-1873 + Conway 1968 + Vol 0 Ch 9 C11 - K3 surface Hodge structure - Niemeier lattices (no-root unimodular Euclidean lattices in $\dim 24$) - Leech lattice + Mathieu M_{24} - 5-family Bridge Object architecture (Vol 0 Ch 9 C11 RIGOROUSLY CLOSED)

Ch 7 Heegner Numbers + Cremona Elliptic Curves (~70%)

Anchor: Stark-Heegner 1952-1967 + Cremona 49a1 + 27a1 + 121a1 + Vol 0 Ch 9 C9 - Stark-Heegner 1967 class-number theorem (9 Heegner discriminants) - Cremona 49a1 = BST canonical elliptic curve $Y^2 = X^3 - 945X - 10206$ - All invariants BST: conductor $49 = g^2$, discriminant $-343 = -g^3$, $j = -15^3 = -(N_c \cdot n_C)^3$ - Vol 0 Ch 9 C9 Stark anchor (T2461 STRUCTURALLY VERIFIED at $\dim_C = 5$)

Ch 8 Monster Moonshine + Supersingular Primes (~70%)

Anchor: Borcherds 1992 + McKay 1979 + Ogg 1975 + Vol 4 Ch 4 T1924 - Borcherds 1992 monstrous moonshine + Conway-Norton 1979 - Ogg 1975 supersingular primes ≤ 71 (= 15 primes) - McKay 1979 McKay-Thompson series - Vol 4 Ch 4 T1924 joint cosmological anchor at $t_{\text{cosmo}} = 47$ (Monster supersingular) - $47 \cdot 59 \cdot 71 = 196883$ (Monster's smallest non-trivial representation dimension)

Ch 9 Cyclotomic $Q(\zeta_N)$ + Reed-Solomon (~65%)

Anchor: K59 RATIFIED + Vol 1 Ch 10 + T2429 RS substrate-tick - Cyclotomic field theory + Galois groups - Reed-Solomon codes over $GF(2^g) = GF(128)$ substrate-tick (T2429 + K59 RATIFIED) - 7-step cyclotomic projection chain (K59 = 7 Mersenne exponent g)

Ch 10 Mersenne Primes + Lucas-Lehmer (~70%)

Anchor: Sub-Substrate Mersenne Hierarchy + T2451 + T2466 + Vol 0 Ch 9 C15 - Mersenne prime theory + Lucas-Lehmer primality test - T2451 Sub-Substrate Mersenne Hierarchy ($\text{rank} \rightarrow N_c \rightarrow g$ via Mersenne map) - T2466 BST Primary Mersenne-Prime Density (6/9 BST primary indices produce Mersenne primes; absorbed into C15) - Vol 0 Ch 9 C15 (Sub-Substrate Mersenne Hierarchy candidate criterion)

Ch 11 Generative Geometry Framing (~40%)

Anchor: Casey Wednesday "D_IV⁵ as emergent TOE" + Vol 0 Ch 9 + Vol 4 Ch 9 - D_IV⁵ as generative geometric object that produces physics - Substrate-cartography reading: "we don't set out to build TOE, yet D_IV⁵ shows us one" - Cross-link to Casey-named #7 D_IV⁵ Rigidity Principle STANDING

Ch 12 Number-Theoretic Foundations Synthesis (~55%)

Anchor: Vol 0 Ch 9 Strong-Uniqueness + Heegner + Monster + Mersenne + cyclotomic unified - Synthesis chapter connecting all NT threads - 7-candidate enumeration C7+C9+C15+C16+C17a+C17b+C18 (Cal #99) cross-link - Multi-level integer-web (BST primary $\rightarrow$ Heegner $\rightarrow$ Mersenne $\rightarrow$ Monster $\rightarrow$ Stark)

~65% existing BST coverage (substantial research; Heegner + Stark + Monster + Mersenne + cyclotomic are mature research lines); 35% to organize at curriculum-readable depth.

Cross-volume reference map

Inputs from Vol 0: D_{IV}^5 + Strong-Uniqueness Theorem 7-candidate enumeration
Inputs from Vol 1: Bergman + Wallach + Faraut-Koranyi anchors + substrate-tick + cyclotomic
Outputs to: Mathematicians joining BST research community; deep-NT cross-reference manual

— Vol 11 Architectural Scaffold v0.1, Saturday 2026-05-23 Wave 2 scaffold